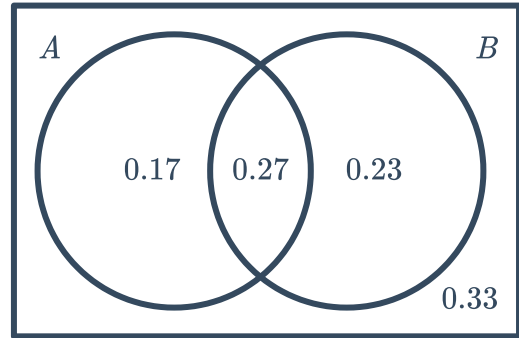


Conditional probability



- 1 Consider the following probability Venn Diagram. Find:

- a $P(A' \cap B)$
- b $P(A' \cup B')$



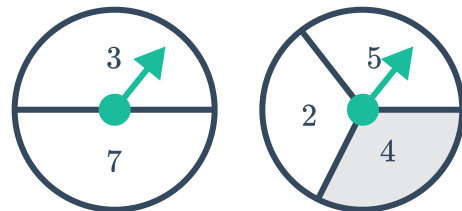
- 2 A and B are two random events with the following probabilities:
Find the value(s) of x if:

- a A and B are mutually exclusive.
- b A and B are independent.

- $P(A) = 0.3 + x$
- $P(B) = 0.2 + x$
- $P(A \cap B) = x$

- 3 The following two spinners are spun and the sum of their respective spins is recorded:

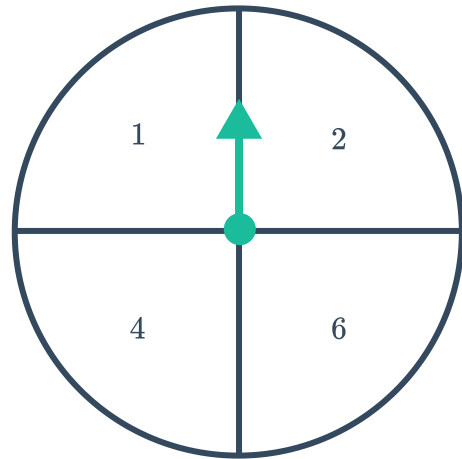
- a Construct a table to represent all possible outcomes.
- b Find probability that a 7 was spun given that the sum is less than 12.
- c Find the probability that the sum is odd given that a 4 was spun.



- 4 The following spinner is spun and a normal 6-sided die is rolled. The product of their respective results is recorded.



- a Construct a table to represent all the possible outcomes.
- b Find the probability that the product was a multiple of 4 given that a 2 was spun.
- c Find the probability that a 6 was rolled given that the product was greater than 10.
- d Find the probability that the product was greater than 4 given that the same number appeared on the dice and the spinner.



Dependent and independent events

- 5 Determine whether the following events of selection are independent or dependent.
- a A card is randomly selected from a normal deck of cards, and then returned to the deck. The deck is shuffled and another card is selected.
 - b The selections of each ball in a lottery.
 - c Two cards are randomly selected from a normal deck of cards without replacement.
 - d Each student is allowed to randomly pick an item from the teacher's prize bag.
- 6 Two events A and B are such that $P(A \cap B) = 0.02$ and $P(A) = 0.2$.
- a Are the events A and B independent if $P(B) = 0.1$?
 - b Are the events A and B mutually exclusive?
- 7 Two events A and B are such that $P(A \cap B) = 0.1$ and $P(A) = 0.5$. Calculate $P(B)$ if events A and B are independent.
- 8 Consider $P(A) = 0.2$ and $P(B) = 0.5$:
- a Find the maximum possible value of $P(A \cup B)$.
 - b Hence, are events A and B mutually exclusive?
 - c Find the minimum possible value of $P(A \cup B)$.



- 9 There are 4 green counters and 8 purple counters in a bag. Find the probability of choosing



- 10** In a game of Blackjack, a player is dealt a hand of two cards from the same standard deck. Find the probability that the hand dealt:
- a** Is a Blackjack. (An Ace paired with 10, Jack, Queen or King.)
 - b** Has a value of 20. (Jack, Queen and King are all worth 10. An Ace is worth 1 or 11.)
- 11** In a game of Draw Poker, a player is dealt a hand of 5 cards from the same deck.
- a** Find the probability of being dealt a:
 - i** Flush (five cards of the same suit)
 - ii** Royal flush (10, Jack, Queen, King and Ace of the same suit)
 - b** How many possible hands are there?
-  **12** In a game of monopoly, two dice are rolled.
- a** Is the outcome of the second die dependent on the outcome of the first?
 - b** If a 5 is rolled on one of the dice, find the probability of rolling a 5 on the second die.
-  **13** In a lottery there are 45 balls.
- a** Find the probability of a particular ball being drawn first.
 - b** A ball is discarded after it has been drawn. If ball number 35 is drawn on the first go, find the probability of ball number 20 being drawn next.
 - c** Is the probability of each successive ball drawn the same as the probability of the first ball drawn?
 - d** Are the draws dependent or independent events?
-  **14** A standard deck of cards is shuffled and placed face down. Dylan attempts to draw five cards randomly so as to get a royal flush (ten, jack, queen, king, ace all of the same suit) .
- a** If he draws a queen of spades on the first go, find the probability that he draws a ten of spades on the second go.
 - b** If he draws a queen of spades on the first go and a ten of spades on the second go, find the probability that he draws a jack of spades on the third go.
 - c** Is the probability of each draw dependent or independent of the previous draw?
- 15** A number game uses a basket with 6 balls, all labelled with numbers from 1 to 6. 2 balls are drawn at random. Find the probability that the ball labelled 3 is picked once if the balls are drawn:
- a** With replacement.
 - b** Without replacement.



- 16** A standard deck of cards is used and 3 cards are drawn out. Find the probability, in fraction form, of the following:
- a** All 3 cards are diamonds if the cards are drawn with replacement.
 - b** All 3 cards are diamonds if the cards are drawn without replacement.
- 17** A number game uses a basket with 9 balls, all labelled with numbers from 1 to 9. 3 balls are drawn at random, without replacement. Find the probability that:
- a** The ball labelled 3 is picked.
 - b** The ball labelled 3 is picked and the ball labelled 6 is also picked.

Tree diagrams and probability trees

- 18** A container holds four counters coloured red, blue, green and yellow. Construct a tree diagram representing all possible outcomes when two draws are done, and the first counter is:
- a** Replaced before the second draw.
 - b** Not replaced before the next draw.
-  **19** A pile of playing cards has 4 diamonds and 3 hearts. A second pile has 2 diamonds and 5 hearts. One card is selected at random from the first pile, then the second.
- a** Construct a probability tree of this situation with the correct probability on each branch.
 - b** Find the probability of selecting two hearts.
-  **20** Two marbles are randomly drawn without replacement from a bag containing 1 blue, 2 red and 3 yellow marbles.
- a** Construct a tree diagram to show the sample space.
 - b** Find the probability of drawing the following:
 - i** A blue marble and a yellow marble, in that order.
 - ii** A red marble and a blue marble, in that order.
 - iii** 2 red marbles.
 - iv** No yellow marbles.
 - v** 2 blue marbles.
 - vi** A yellow marble and a red marble, in that order.
 - vii** A yellow and a red marble, in any order.



- 21** Each year at Sicily High School, students take part in a competition to recite the digits in π . Maria won in 2014 and 2015, but there's a new contender, Dario.
- If only these two compete, construct a tree diagram of the possible winners from 2014 to 2018 inclusive.
 - If Maria and Dario are equally likely to win each year, find the probability that Dario wins in 2016 given that Maria won in 2018.
 - Find the probability that Dario wins two of the three competitions from 2016 to 2018 given that Maria won in 2017.
- 22** When Fred gets ready for work in summer, he first decides whether it will be a hot day (H) or not (N) and then decides whether to wear a tie (T) or to just dress casually (C). The chance of Fred deciding it will be a hot day is 0.7. If he decides it will be a hot day, there is a 0.2 chance he will wear a tie. If he decides it will not be a hot day, there is a 0.85 chance he will wear a tie.
- Construct a probability tree for this situation.
 - Calculate the probability Fred decided it was a hot day given that he wears a tie. Round your answer to two decimal places.
- 23** In tennis if the first serve is a fault (out or in the net), the player takes a second serve. A player serves with the following probabilities:
- First serve in: 0.55
 - Second serve in: 0.81
- Construct a probability tree showing the probability of the first two serves either being in or a fault.
 - Find the probability that the player needs to make a second serve.
 - Find the probability that the player makes a double fault (both serves are a fault).
-  **24** Han draws two cards, without replacement, from a set of cards numbered 1 to 4 to create a 2-digit number. If he draws a 1 then a 2 the number 12 is formed.
- Construct a tree diagram to find all possible outcomes.
 - Find the probability that the number formed is:
 - Odd
 - Divisible by 11.
 - Even
 - Divisible by 3.
 - Divisible by 5.
 - Divisible by 7.
 - Divisible by 4.

25 Nine pilots from StarJet and 7 pilots from  offer to take part in a rescue operation. If 2 pilots are selected at random:

- a** Construct a probability tree showing all possible combinations of airlines from which the pilots are selected.
- b** Find the probability that the two pilots selected are from:
 - i** The same airline
 - ii** Different airlines

26 Sophia has three races to swim at her school swimming carnival. The probability that she wins a particular race is dependent on whether she won the previous races, as summarised below:

- The chance she wins the first race is 0.7.
 - If she wins the first race the chance of winning the second is 0.8.
 - If she loses the first race then the chance of winning the second is 0.4.
 - If she wins the first two then the chance of winning the third race is 0.9.
 - If she lost the first two then her chance of losing the third race is 0.9.
 - If she won only one of the first two races, then the chance of winning the third is 0.6.
- a** Construct a probability tree to represent all outcomes in this situation.
 - b** Calculate the probability Sophia won all three races correct to three decimal places.
 - c** Calculate the probability Sophia won the third race, correct to three decimal places.